LEVETIRACETAM

Few interactions

Mood & behaviour changes

CHEMICAL NAME (2S)-2-(2-oxopyrrolidin-1-yl)butanamide
CLASS Pyrrolidone antiepileptic (SV2A ligand)
TRADE NAMES Keppra · Levetiracetam (generic) · Elepsia XR

THERAPEUTIC USES / INDICATIONS

- Focal seizures (monotherapy or adjunct); myoclonic and generalised tonic-clonic seizures (adjunct).
- Status epilepticus (second-line IV).
- Seizure prophylaxis (neurosurgery, ICU); first-choice in many settings due to safety.

PHARMACODYNAMICS

- Binds **synaptic vesicle protein 2A (SV2A)** → modulates neurotransmitter release; inhibits N-type calcium channels.

ADMINISTRATION

- **PO** twice daily with or without food.
- **IV** infusion over 15 min (diluted); IV dose = oral dose.
- Can be given via NG tube (oral solution).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Somnolence, fatigue, dizziness.
- Headache.
- Irritability, aggression, mood changes (especially children).

ADVERSE EFFECTS (SERIOUS)

- Depression, psychosis, suicidal ideation.
- SJS/TEN (rare).
- Pancytopenia (rare).

DOSAGES

- **Adult:** start 500 mg twice daily, increase by 500 mg BD every 2 weeks; max 1.5 g twice daily.
- **Status epilepticus:** 60 mg/kg IV (max 4500 mg; ESETT) over 10–15 min.
- **Children:** 10 mg/kg twice daily, up to 30 mg/kg twice daily.
- **Renal impairment:** reduce per CrCl.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** renal impairment, psychiatric history.

KEY INTERACTIONS

- No significant pharmacokinetic interactions (doesn't affect contraceptives).

PHARMACOKINETICS

- **Peak:** 1 h. Bioavailability ~100%.
- Not CYP-metabolised (enzymatic hydrolysis); minimal protein binding.
- Excreted in urine (66% unchanged). **t½:** 6–8 h.

NURSING MANAGEMENT

- Adjust dose for renal function.
- Monitor mood and behaviour; involve family.
- Seizure chart.
- IV: dilute and infuse over 15 min.

PATIENT EDUCATION

- Report mood changes, irritability, low mood or thoughts of self-harm.
- May cause drowsiness — avoid driving until seizure-free per law.
- Do not stop suddenly.

KEY POINTS

- SV2A ligand; broad, safe, few interactions.
- Status: 60 mg/kg IV.
- 500 mg BD, up to 1.5 g BD.
- Behaviour/mood changes; renal dosing.

SOURCES CHECKED **[732]** Levetiracetam 1000 mg film-coated tablets (UK SmPC) **[733]** Levetiracetam 100 mg/ml concentrate for solution for infusion (UK SmPC) **[725]** Kapur et al., N Engl J Med 2019 **[734]** Levetiracetam, oral (US label) **[735]** Levetiracetam, IV (US label)
Full references p. 432–457